

Adora PINN — Step 2: Starting Atmosphere

As-built notes · Ivan Milic · 2026-09-20 · step 2 of pinn/plan.md

Step 2 builds the warm-start atmosphere: a lightweaver FAL-C column resampled to a regular z grid and cloned into a 128×128 map. Builder [pinn/make_start_atmosphere.py](#); check [pinn/step2_atmosphere.ipynb](#). Saved (SI) to [/dat/milic/adora.pinn_develop/test_atm_falc_128.npz](#).

z convention (resolved). This lightweaver FAL-C has $z \in [0, 2342]$ km with $z = 0$ at the deep sub-photosphere and $\tau=1$ near $z \approx 100$ km (*not* $z=0$ at $\tau=1$). Per your call, we relabel $z' = z_{\text{lw}} - 100$ so the model bottom is at $z' = -100$ km and sample FAL-C over $z_{\text{lw}} \in [0, 1100]$ km — entirely within the model, so **no extrapolation**; $\tau=1$ lands near $z' \approx 0$. Grid: $z' \in [-100, 1000]$ km, 20 km steps, **56 points**, cloned to 128^2 .

Channels (SI, the 7 PINN outputs). T [K], P [Pa], v_z [m/s], $B_{x,y,z}$ [T], v_{turb} [m/s]. T, P, v_{turb} are the cloned FAL-C column ($P = (n_{\text{H}}\Sigma_A + n_e)k_B T$, $\Sigma_A=1.095$); $B=0$ (non-magnetic). A smooth **velocity structure** is added so $v_z \neq 0$: $v_z = 3 \text{ km/s} \times \cos(2\pi 3x) \cos(2\pi 2y) \times e^{-(z'/400 \text{ km})^2}$ — a granulation-like horizontal pattern with a photospheric envelope, giving the field genuine (x, y, z) structure to fit in step 3. Reference $n_e, n_{\text{H,tot}}$ are also stored (1-D) for cross-checking the EOS, which will supply $n_e, n_{\text{H,tot}}$ from (T, P) downstream.

Profile (sanity). Standard FAL-C shape: $T = 4502\text{--}9400$ K with the temperature minimum at $z' \approx 540$ km; P monotonic $3.75\text{--}1.94 \times 10^4$ Pa; $\tau=1$ point ($z' \approx 0$) at $T \approx 6600$ K. Only v_z varies horizontally (± 3 km/s); T, P, v_{turb}, B are cloned (std across $x, y = 0$).

Notes. [/data/milic](#) did not exist, so the file went to your established [/dat/milic](#). The cube is ~ 51 MB ($56 \times 128 \times 128$, 7 channels). **Next:** step 3 — pretrain the atmosphere field (step 1) to this cube for a warm start. Files uncommitted.